STATION: SALEM

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions For A Transfer To Cold Leg Recirculation

TASK NUMBER: 1150030501

JPM NUMBER: 13-01 NRC Sim b

ALTERNATE PATH: [X]

K/A NUMBER: E011 EA1.11

IMPORTANCE FACTOR: 4.2 (**RO**) 4.2 (**SRO**)

APPLICABILITY:
EO [ ] **RO** [X] **STA** [ ] **SRO** [X]

EVALUATION SETTING/METHOD: Simulator

REFERENCES: 2-EOP-LOCA-3 Rev. 29 (Rev. checked 10-7-14)

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 12 min

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: ____________

Developed By: **G Gauding**
Instructor
Date: **9-6-14**

Validated By: **D Tait**
SME or Instructor
Date: **10-15-14**

Approved By: J J Carey Carney
Training Department
Date: 10-23-14

Approved By: (MOC)
Operations Department
Date: 10-23-14

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY: ____________

GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

NAME: ______________________

DATE: ______________________

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions For A Transfer To Cold Leg Recirculation

TASK NUMBER: 115 010 05 01

SIMULATOR SETUP:
IC- 252
MALF: VL0121 22SJ44 fails to 0% open
REMOTES: SW27D 22 SW pump control power OFF
SW43D 26 SW pump control power OFF
I/O: **A908 OVDI** 24 SW pump start PB OFF
A909 OVDI 25 SW pump start PB OFF
A403 OVDI 22 CS pump stop PB OFF
A614 OVAO 22 CCHX Outlet Temperature 200

- **Place Bezel Cover on 22 SW pump**

INITIAL CONDITIONS:

- A LBLOCA has occurred on 22 RC Loop.
- All Vital Buses are energized from off-site power.
- Operators are performing actions in 2-EOP-LOCA-1, Loss of Reactor Coolant.
- 22 SW pump is C/T.
- 26 SW pump tripped 2 minutes ago.
- RWST lo level alarm has just annunciated.

INITIATING CUE:

You are the RO. Perform 2-EOP-LOCA-3, Transfer to Cold Leg Recirculation, beginning with Step 1.

Successful Completion Criteria:
1. **All critical steps completed.**
2. **All sequential steps completed in order.**
3. **All time-critical steps completed within allotted time.**
4. **JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.**

Task Standard for Successful Completion:

1. Transfer to CL recirc with single train operation.

TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: ____________________
DATE: ____________________

SYSTEM: Residual Heat Removal
TASK: Align RHR Suction to Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		START TIME: ____________			
	1	Do not implement any FRPs until directed by this EOP.	Reads step.		
	2	Is “Cont Sump Ch A(B) Level >62%” lit	Checks Cont Sump Ch A(B) Level >62% lights and answers YES.		
*	3	Depress “SUMP AUTO ARMED” pushbuttons on 21 and 22 SJ44 bezels	Depresses “SUMP AUTO ARMED” pushbuttons on 21 and 22 SJ44 bezels.		
*	4	Remove lockouts for the following: • 2SJ67 (SI Pumps Miniflow) • 2SJ68 (SI Pumps Miniflow) • 2SJ69 (Common Suction)	Rotates lockout switches on 2RP4 to the VALVE OPERABLE position for: • 2SJ67 (SI Pumps Miniflow) • 2SJ68 (SI Pumps Miniflow) • 2SJ69 (Common Suction)		
	5	Are 21 and 22 SJ44 (Sump Valves) open?	Determines Sump Valve 21SJ44 is open and Sump Valve 22SJ44 is shut.		
	5.1	Reset SI	Determines SI is reset.		
	5.1	Reset Emergency Loading for each SEC	Determines Emergency Loading for each SEC is reset.		
	5.2	Is 21SJ44 open	Determines 21SJ44 is open.		
*	5.2	Stop 22 RHR pump	Depresses stop PB for 22 RHR pump.		

TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: ______________________
DATE: ______________________

SYSTEM: Residual Heat Removal

TASK: Align RHR Suction to Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.2	Close 2SJ69 (Common Suction)	Depresses close PB for Close 2SJ69 (Common Suction) and verifies green close light illuminates.		
	5.2	Start 21 RHR pump.	Determines 21 RHR pump is running.		
	5.2	Initiate close on 22RH4 (Pump Suction) and continue.	Depresses close PB for 22RH4 (Pump Suction) and continues.		
	5.2	Initiate open on 22SJ44 and continue.	Depresses open PB for 22SJ44 and continues.		
	5.3	When 22SJ44 opens, Then start 22 RHR pump.	Does not start 22 RHR pump **Note**: 22SJ44 will not open.		
	6	IF Blackout loading occurs on any vital bus after SI reset, then perform actions per Table A.	Reads step.		
	7	Reset SI	Determines SI is reset on Trains A and B.		
		Reset Emergency loading for each SEC.	Determines Emergency Loading is Reset for each SEC.		
		Are all SECs reset?	Determines all SECs are reset.		
		Reset 230V control centers.	Determines all 230V control centers are reset.		
	8	Are both CS pumps running	Determines both CS pumps are running.		

TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: ______________________
DATE: ______________________

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TASK: Align RHR Suction to Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Stop 22 CS pump	Depresses stop PB for 22 CS pump and determines pump does not stop. **Note:** May dispatch an operator to locally open or standby to open 22 CS pump breaker.		
		Is 22 CS pump stopped	Determines 22 CS pump is not stopped.		
*		Stop 21 CS pump	Depresses stop PB for 21 CS pump and Determines pump has stopped.		
*	9	Close 21 and 22RH19 (RHR HX Disch x-conn valves)	Depresses close PB for 21 and 22RH19 (RHR HX Disch x-conn valves) and verifies shut lights illuminate.		
		Stop 23 charging pump	Depresses stop PB for 23 charging pump and verifies it stops.		
	10	Select appropriate flowpath transition step from Table B Go to selected step	Selects flowpath transition step 11 with all vital buses energized. Goes to Step 11.		
	11	Is <u>Any</u> 4KV vital bus energized by DG	Determines no 4KV vital bus energized by DG.		
	11.1	Are <u>at least three</u> SW pumps running	Determines only 2 SW pumps are running. **Note**: If operator attempts to start 24 or 25 SW pumps they will not start.		
*		Stop 2 CFCUs	Depresses stop PBs for 2 running CFCUs and verifies they stop.		

TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: ____________________

DATE: ____________________

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Go to Step 118	Goes to Step 118.		
*	118	Stop the following pumps: • 22 SI pump • 21 Charging pump • 22 AFW pump	Depresses stop PBs for: • 22 SI pump • 21 Charging pump • 22 AFW pump and verifies they stop. **Note:** If SG lo lvl is present 22 AFW pp will not stop.		
		Is 22 CCW HX in service	Determines that 22 CCW HX is **not** available, **nor** being returned to service, based on high temperature on 2CC2. (Temp is ~196°F)		
		Open 21CC16 (CC Supply to RHR HX Valve)	Determines 21CC16 (CC Supply to RHR HX Valve) is open.		
		Start 21 RHR pump	Determines 21 RHR pump is in service		
		Stop 22 RHR pump	Determines 22 RHR pump is stopped.		

TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: ____________________
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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*		Close 22CC16 (CC Supply to RHR HX Valve)	Depresses 22CC16 (CC Supply to RHR HX Valve) close PB and verifies it shuts.		
		Close 2SJ68 and 2SJ67(SI Pumps Miniflow Valves)	Depresses close PB for Close 2SJ68 and 2SJ67(SI Pumps Miniflow Valves) and verifies they shut.		
		Close 2RH1 <u>AND</u> 2RH2 (Common Suction Valves)	Determines 2RH1 <u>AND</u> 2RH2 (Common Suction Valves) are shut.		
	119	Is 21 RHR pump running	Determines 21 RHR pump is running.		
	120	Open 21SJ45 (RHR Discharge to SI pump valve)	Depresses open PB for 21SJ45 (RHR Discharge to SI pump valve) and verifies it opens.		
		Terminate JPM			

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

Ø ___ 1. Task description and number, JPM description and number are identified.

Ø ___ 2. Knowledge and Abilities (K/A) references are included.

Ø ___ 3. Performance location specified. (in-plant, control room, or simulator)

Ø ___ 4. Initial setup conditions are identified.

Ø ___ 5. Initiating and terminating Cues are properly identified.

Ø ___ 6. Task standards identified and verified by SME review.

Ø ___ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

Ø ___ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 29 Date 10/15/14

Ø ___ 9. Pilot test the JPM:
   a. verify Cues both verbal and visual are free of conflict, and
   b. ensure performance time is accurate.

_______ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

_______ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [signature] Date: 10/15/14

SME/Instructor: ____________________ Date: ____________

SME/Instructor: ____________________ Date: ____________

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